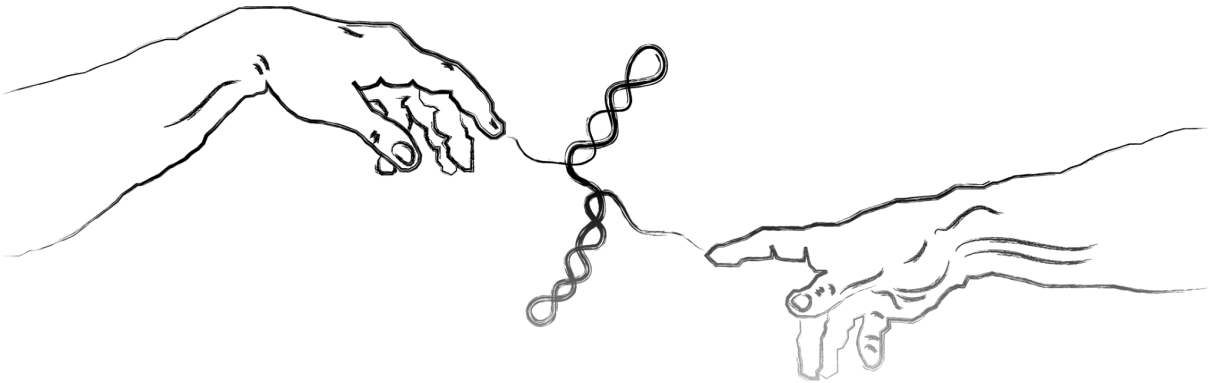

Slot 4: SHAPE

*Methods for investigating the **RNA structurome** and **RNA-RNA interactome** W24/25, by Professor Irmtraud Meyer*

Students: Baptiste Roizard, Hang Mai Anh Vo
Department of Mathematics and Computer Science, Freie Universität Berlin

Illustration taken from here: https://github.com/dincarnato/RNAFramework

Slot 4: SHAPE Directed Discovery of New Functions in Large RNAs

ACCOUNTS of chemical research

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Article

SHAPE Directed Discovery of New Functions in Large RNAs

Kevin M. Weeks*

Cite This: *Acc. Chem. Res.* 2021, 54, 2502–2517

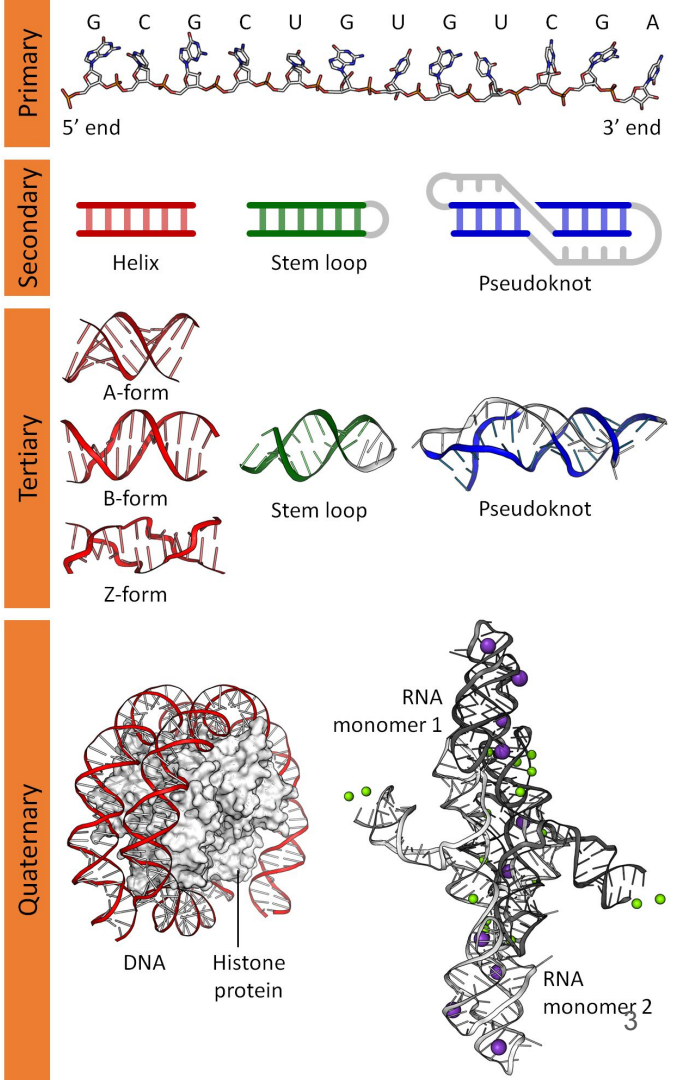
Read Online

Introduction: Why RNA structure matters

RNA carries genetic information

RNA folds into complex shapes that regulate function

RNA structure affects translation, splicing, and regulation



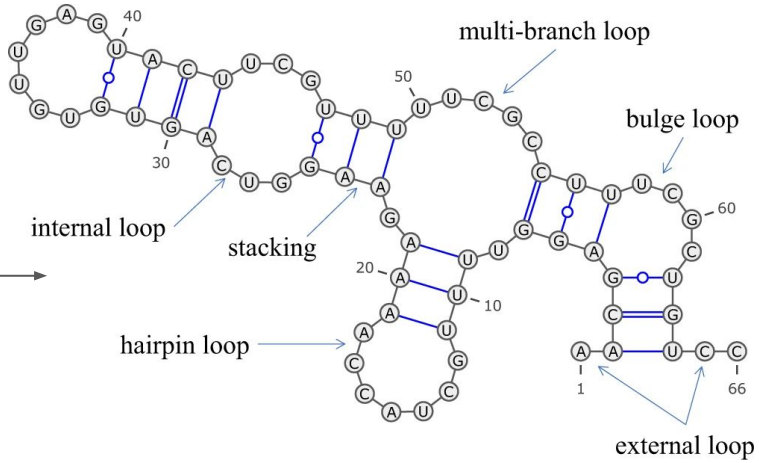
Thomas Shafee (2017). Summary of nucleic acid structure (primary, secondary, tertiary, and quaternary) using DNA helices and examples from the VS ribozyme and telomerase and nucleosome. *Wikipedia*, *Nucleic acid structure*.https://en.wikipedia.org/wiki/Nucleic_acid_structure

Motivation for the study

Historical Challenges in RNA Structure Study

Limited methods to probe RNA at nucleotide resolution

Early models depicted RNA as a ‘random coil’

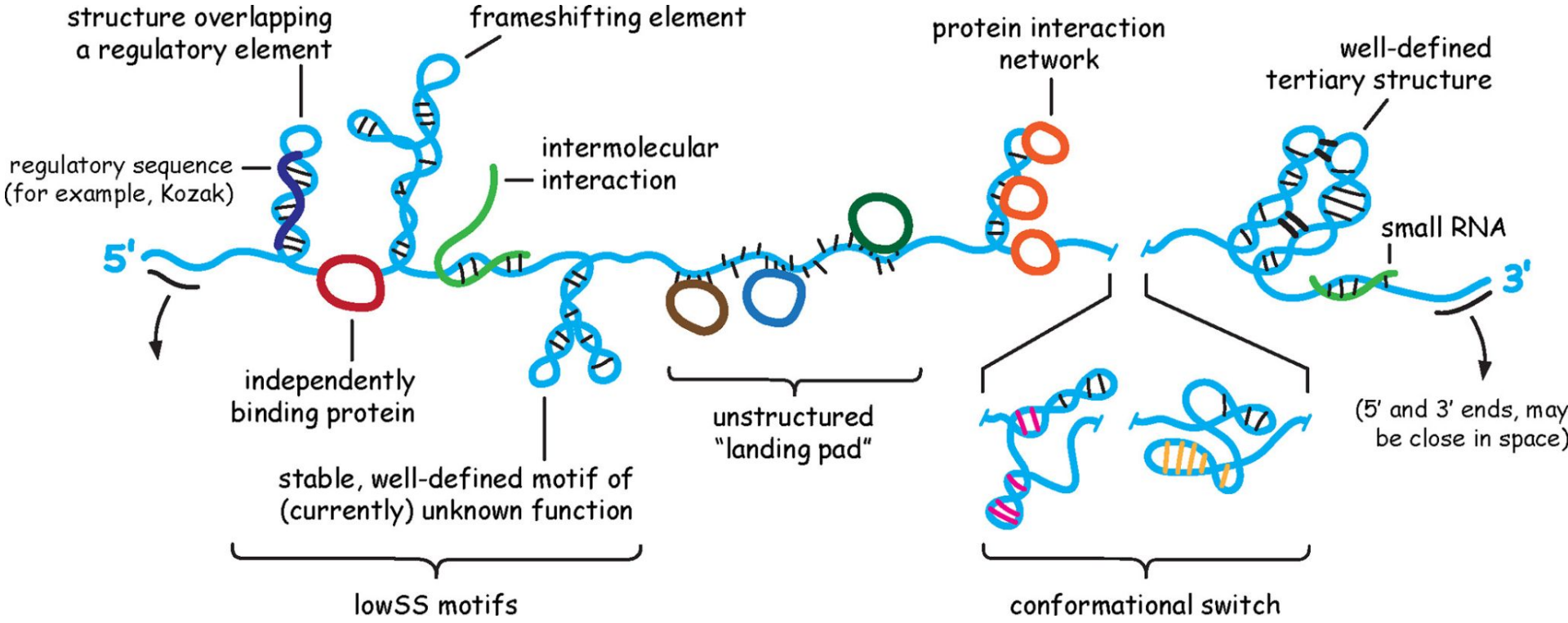


What SHAPE Aims to Solve

Need for large-scale, quantitative RNA structure analysis

Sato, K., Akiyama, M., & Sakakibara, Y. (2021). RNA secondary structure prediction using deep learning with thermodynamic integration. *Nature communications*, *12*(1), 941.

Architecture and potential internal structures of RNA



Introduction to SHAPE chemistry

What is SHAPE?

Selective 2′-**H**ydroxyl **A**cylation analyzed by **P**rimer **E**xtension

Measures nucleotide flexibility, indirectly revealing structure

SHAPE Mechanism

Chemical reagent reacts with ***flexible nucleotides***

Modified sites detected via reverse transcription (RT-STOP)

Introduction to SHAPE chemistry

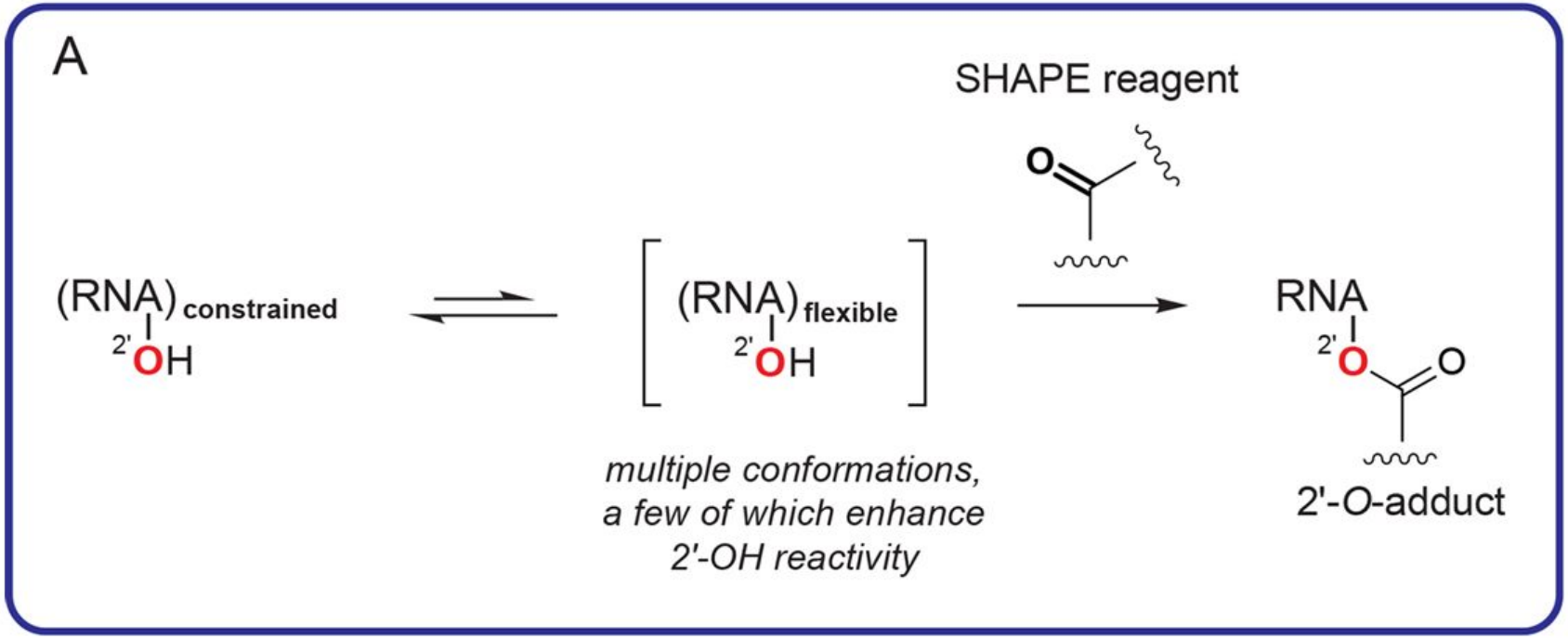
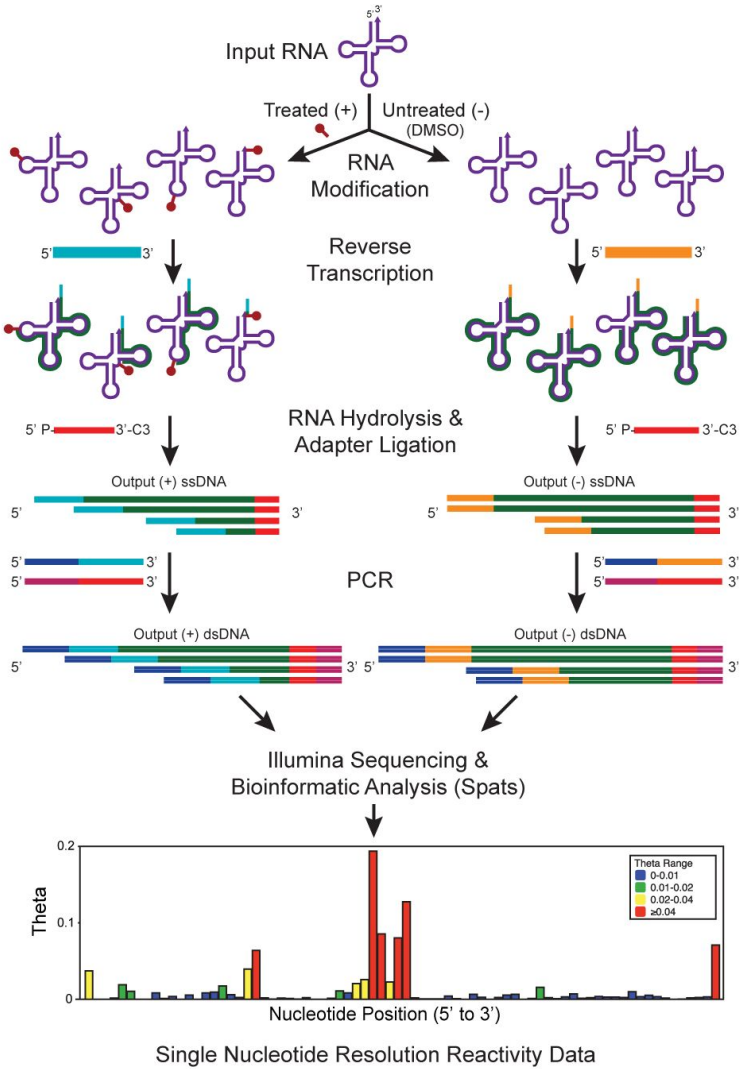
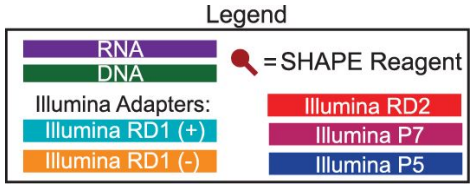


Diagram of how SHAPE works

The basic SHAPE-Seq protocol



Loughrey, D., Watters, K. E., Settle, A. H., & Lucks, J. B. (2014). SHAPE-Seq 2.0: systematic optimization and extension of high-throughput chemical probing of RNA secondary structure with next generation sequencing. *Nucleic acids research*, *42*(21), e165-e165.



SHAPE-MaP – A Major Advancement

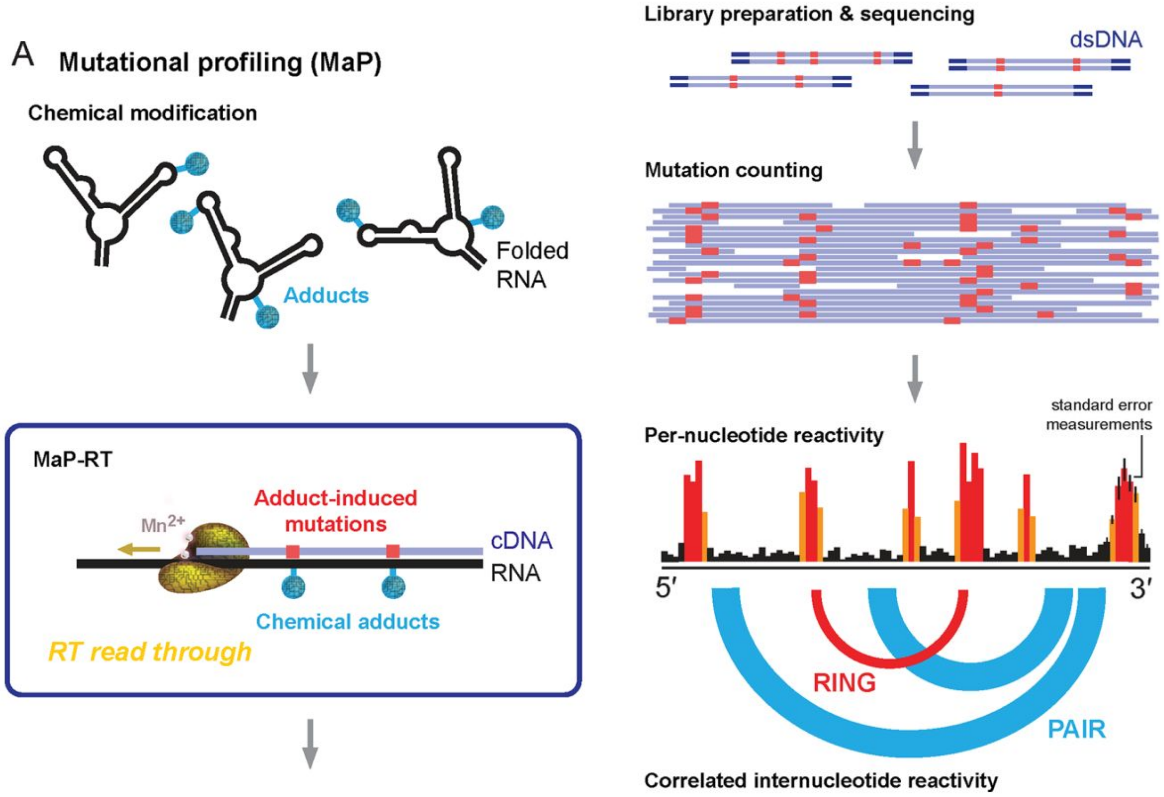
Mutational Profiling (MaP) Improves SHAPE

- Traditional methods relied on RT stops
- SHAPE-MaP allows **mutation-based detection** (more precise)

Advantages of SHAPE-MaP

- Probes **entire RNA molecules**
- Works in **living cells**
- Provides **quantitative data (standard errors)**

Mutational Profiling (MaP) Workflow



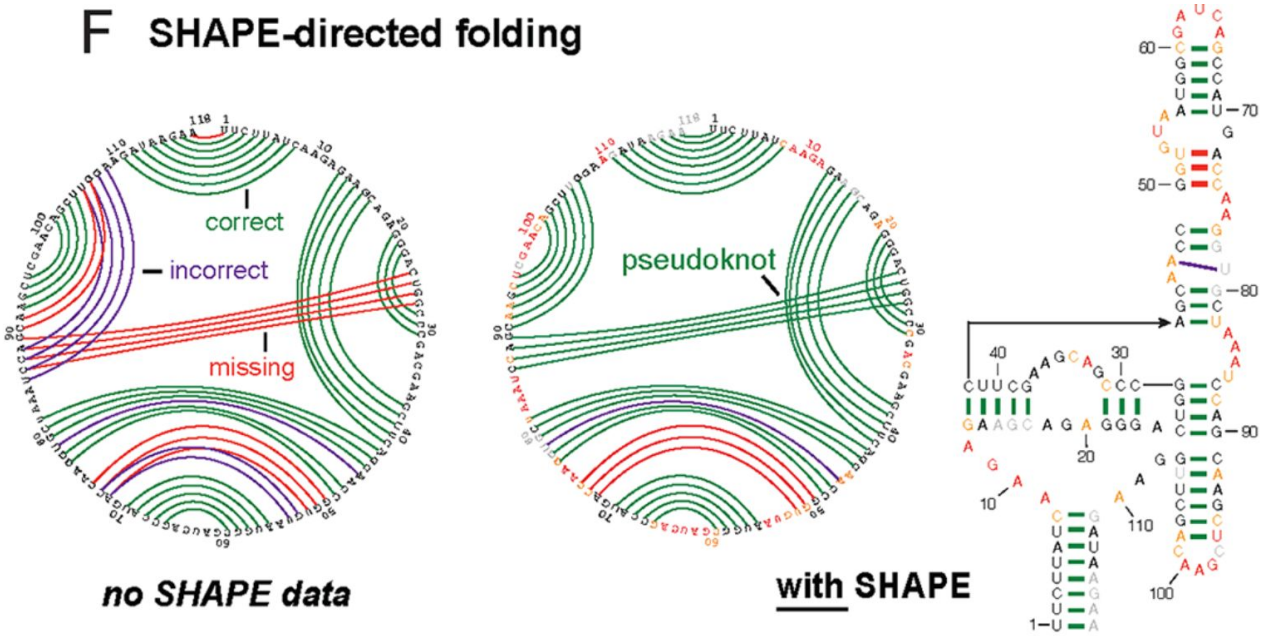
- RNA is treated with a **structure-sensitive reagent**, modifying flexible nucleotides.
- **Reverse transcription (RT)** incorporates mutations at modified sites instead of stopping.
- **Sequencing and alignment** of cDNA reads generate SHAPE reactivity profiles.
- Reactivity profiles reveal **RNA secondary and tertiary structures**.
- Data can be analyzed at the **nucleotide level or through internucleotide correlations**.

SHAPE vs. SHAPE-MaP Workflow

- **SHAPE (Selective 2′-Hydroxyl Acylation analyzed by Primer Extension):**
  - RNA is treated with a SHAPE reagent that modifies flexible nucleotides.
  - Reverse transcription stops at modification sites.
  - Gel or capillary electrophoresis is used to detect truncations, generating reactivity profiles.
- **SHAPE-MaP (Mutational Profiling with SHAPE):**
  - RNA is treated with a SHAPE reagent, introducing modifications.
  - Reverse transcription incorporates mutations instead of stopping.
  - cDNA is sequenced, and reactivity profiles are inferred from mutation rates.

⇒ SHAPE relies on primer extension stops, whereas SHAPE-MaP uses mutational signatures in full-length cDNA for more robust and quantitative structural analysis.

SHAPE reveals RNA's structural complexity



Comparison of SAM-I riboswitch secondary structure prediction:

- **Without SHAPE data (left)**: Less accurate secondary structure prediction.
- **With SHAPE data (right)**: Significantly improved accuracy, especially for complex RNAs like those with pseudoknots.

Arcs in the structure:

- **Green**: Correct base pairs.
- **Purple**: Incorrect base pairs.
- **Red**: Missing base pairs relative to the accepted structure.

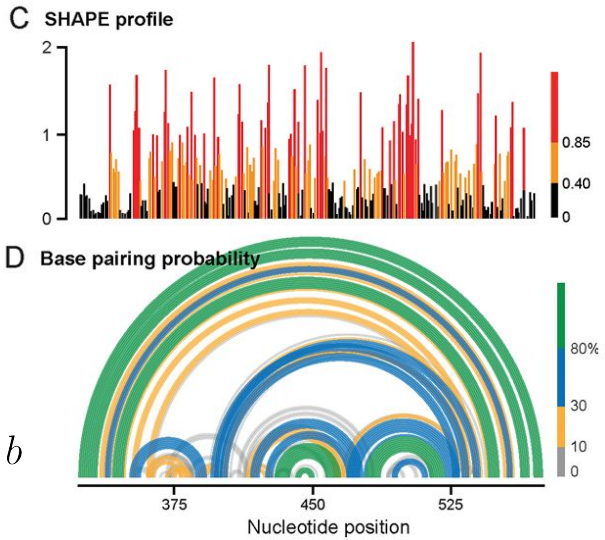
SHAPE Data Interpretation & Statistical Analysis

How SHAPE Reactivity is Processed

Reactivity values range from **0 (rigid)** to **high (flexible)**

Data normalized and used to infer RNA secondary structures

Visualisation: probability arcs



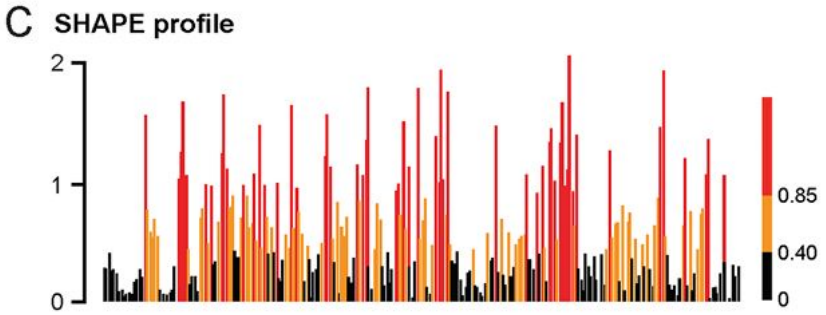
Mathematical Model for SHAPE-Guided Folding

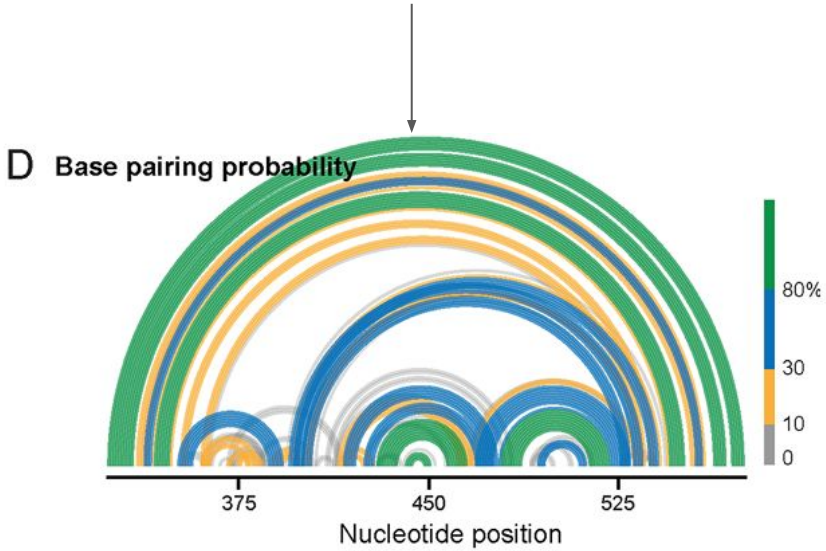
SHAPE pseudo-free energy term: $\Delta G_{\mathrm{SHAPE},i} = m \ln(\mathrm{SHAPE}_i + 1) + b$

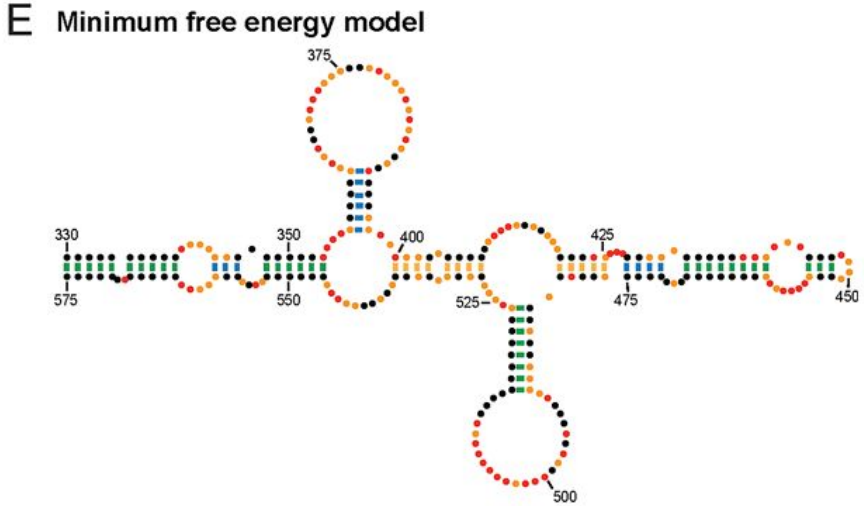
m is a penalty for base pairing that increases with SHAPE reactivity

b reflects a favorable term for pairing at nucleotides with low SHAPE reactivities

How SHAPE data shapes the prediction of secondary structure







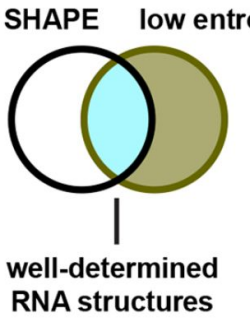
Finding Functional RNA Elements Using SHAPE

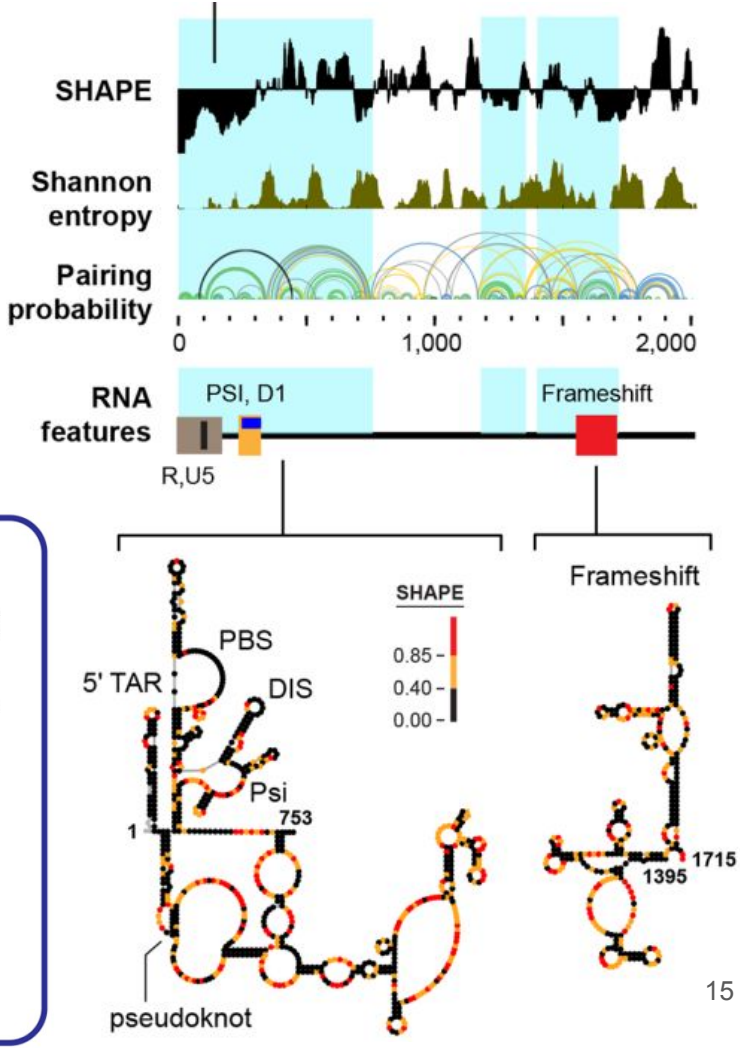
Low SHAPE - Low Shannon Entropy (LowSS) Regions

Identifies well-determined RNA structures

Case Study: HIV-1 Frameshift Element

$$H_i = -\sum_{j=1}^{J} p_{i,j} \log_{10} p_{i,j}$$



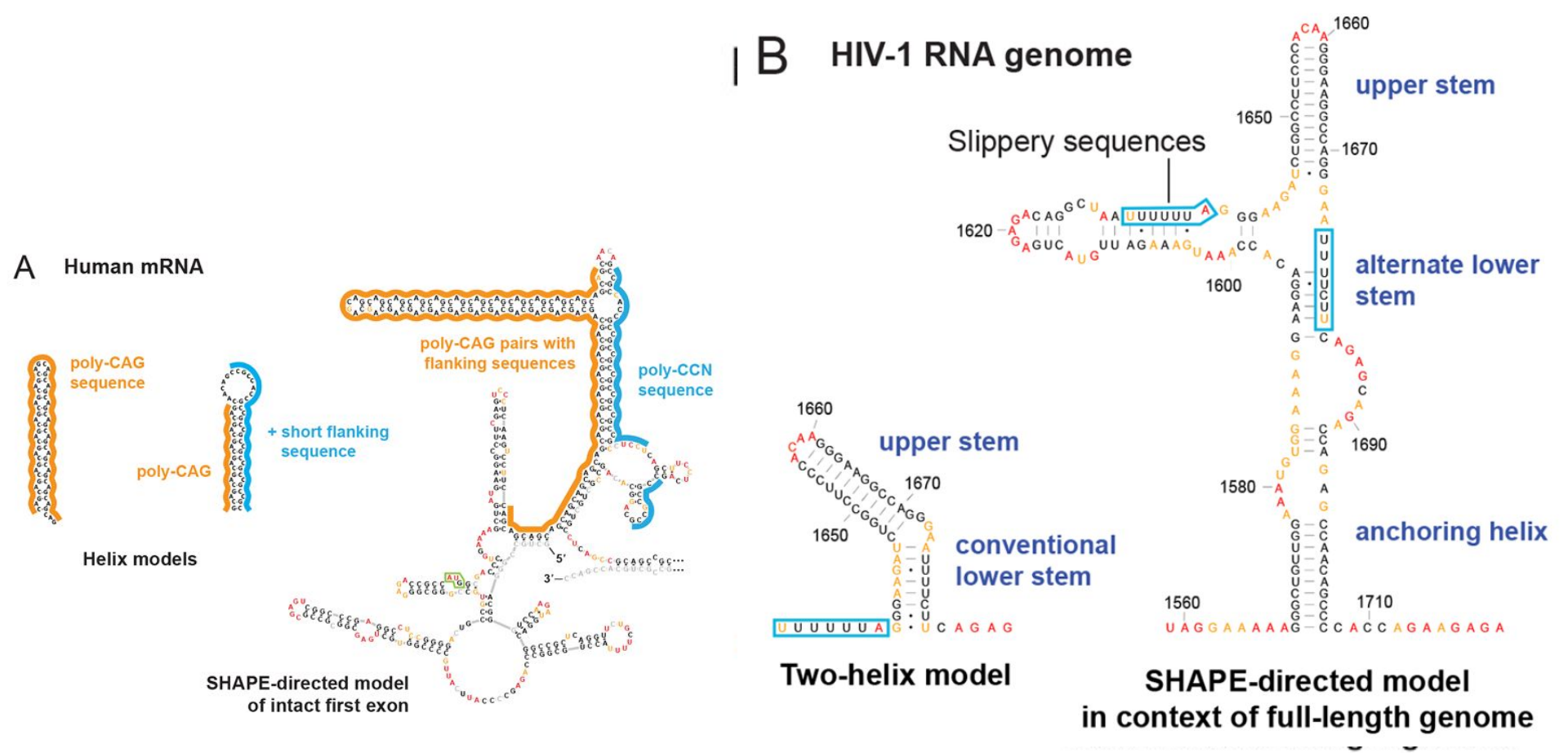


SHAPE and Shannon entropy

A practical way to define **LowSS regions in SHAPE-informed RNA structure prediction** is:

$$\text{LowSS} = \{i \mid \text{SHAPE}_i \leq 0.3 \text{ and } S(i) \leq 0.3 \text{ bits}\}$$

- **If SHAPE ≤ 0.3 and Shannon entropy ≤ 0.3 → Highly structured, LowSS region.**
- **If SHAPE ≤ 0.3 but entropy > 0.3** → Structure may be stable, but some **alternative conformations exist**.
- **If SHAPE > 0.3 and entropy > 0.5** → HighSS region (flexible or dynamic).

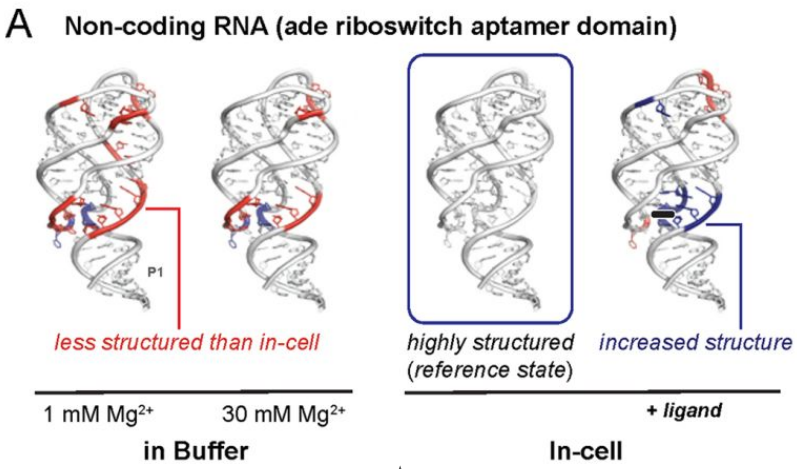

A
Human mRNA
poly-CAG sequence
poly-CAG
+ short flanking sequence
Helix models
poly-CAG pairs with flanking sequences
poly-CCN sequence
5'
3'
SHAPE-directed model of intact first exon
B
HIV-1 RNA genome
Slippery sequences
upper stem
alternate lower stem
anchoring helix
1660
1650
1670
1620
1600
1690
1580
1560
1710
upper stem
conventional lower stem
Two-helix model
SHAPE-directed model in context of full-length genome

RNA Structure in Living Cells

Challenges of Probing RNA in Cells

Cellular environment affects RNA structure

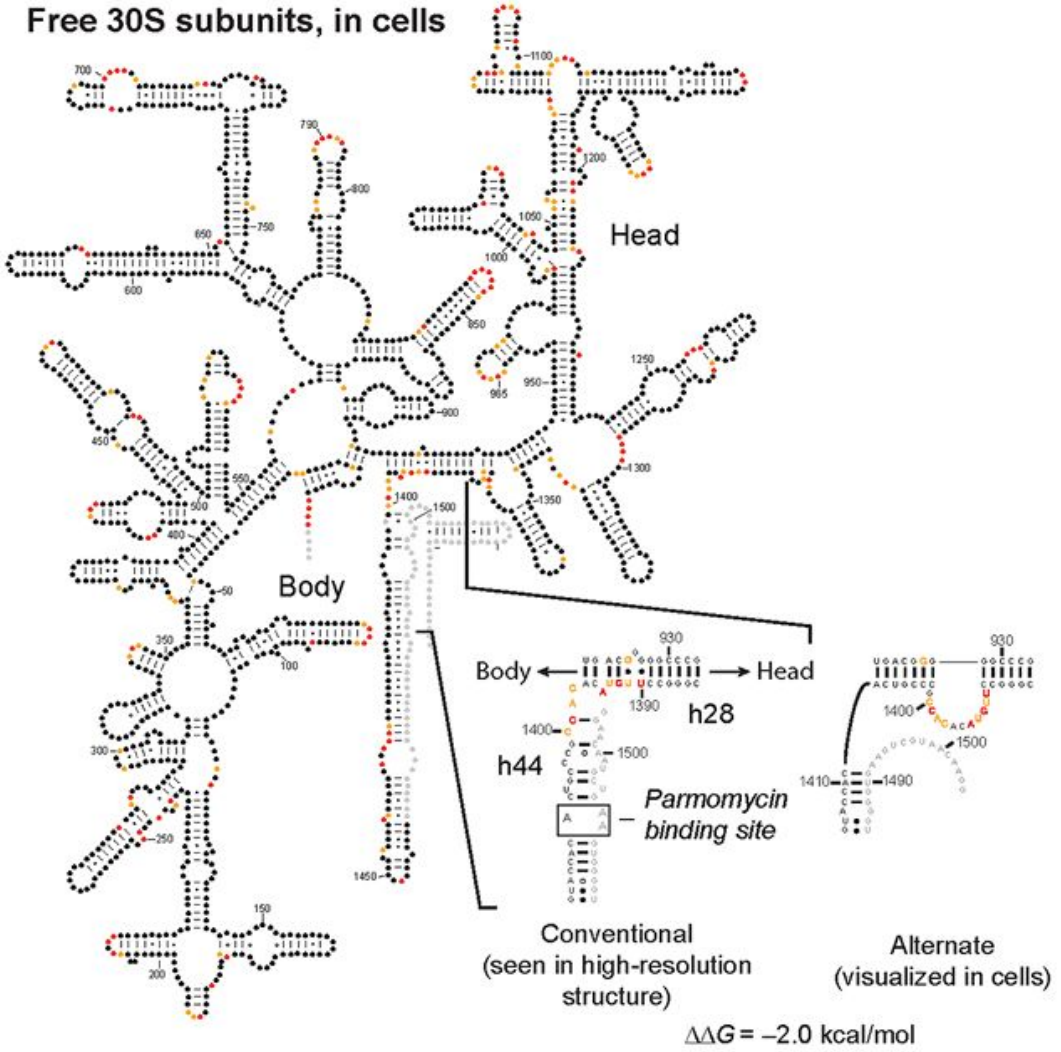
Differences between in vitro and in vivo SHAPE data

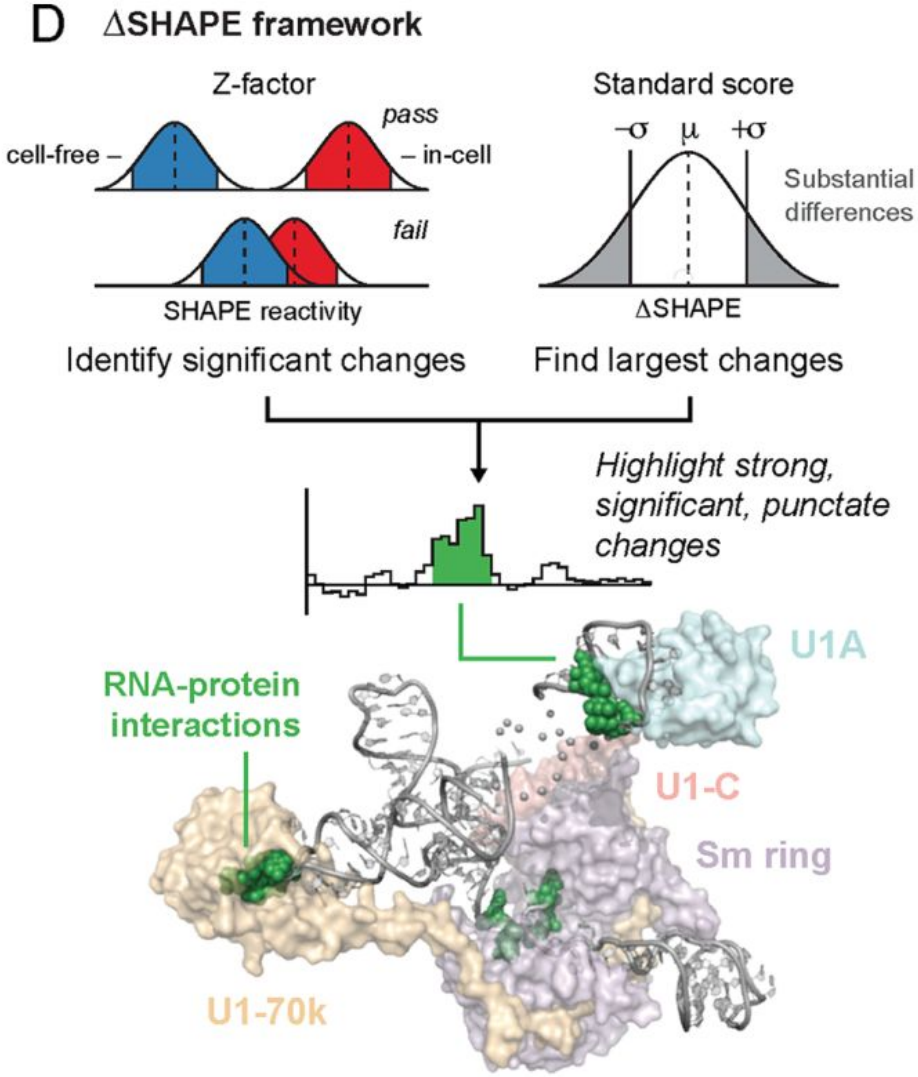


In-Cell SHAPE Applications

Riboswitch RNA is much more highly structured in cells in E. coli

Xist (X-inactive specific transcript) lncRNA structural mapping: role in X chromosome inactivation (XCI)


B
Free 30S subunits, in cells
Head
Body
Body
Head
930
1390
h28
1400
1500
h44
Parmomycin
binding site
930
1400
1500
1410
1490
Conventional
(seen in high-resolution
structure)
Alternate
(visualized in cells)
ΔΔG = –2.0 kcal/mol


D ΔSHAPE framework
Z-factor
pass
cell-free –
– in-cell
fail
SHAPE reactivity
Standard score
−σ μ +σ
Substantial differences
ΔSHAPE
Identify significant changes
Find largest changes
Highlight strong, significant, punctate changes
U1A
RNA-protein interactions
U1-C
Sm ring
U1-70k

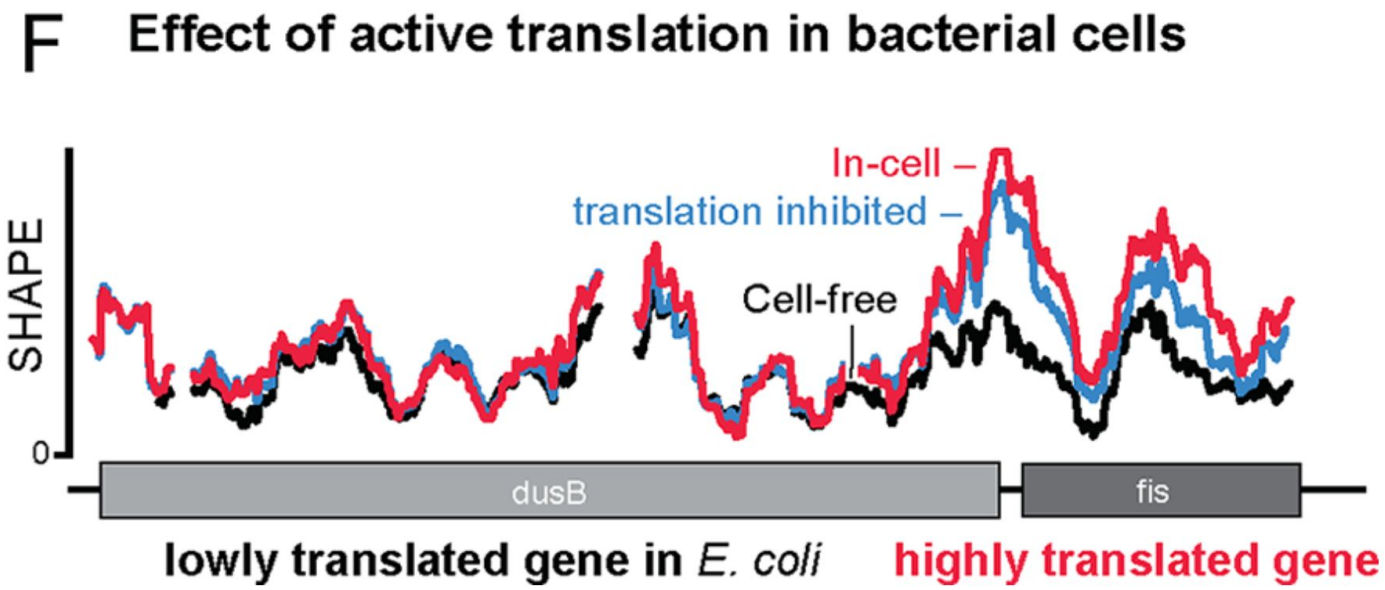
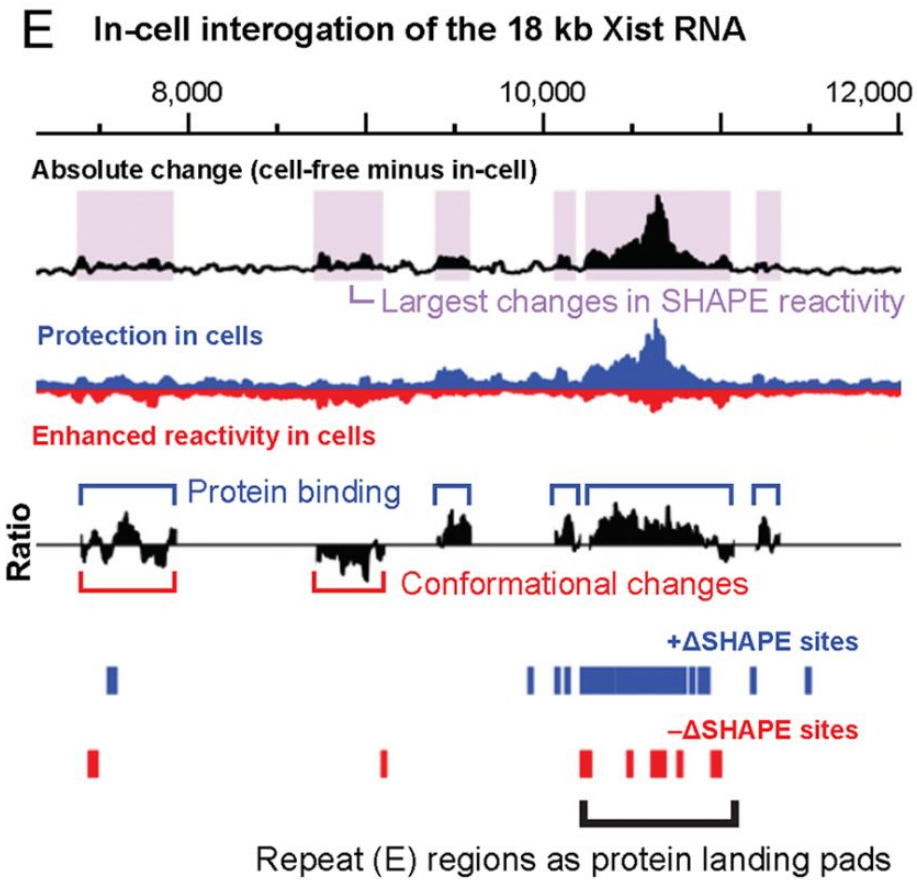


Figure 6. (F) Effect of translation on RNA structure in E. coli cells. SHAPE reactivities increase, relative to the cell-free state, specifically in highly translated genes.


E
In-cell interogation of the 18 kb Xist RNA
8,000
10,000
12,000
Absolute change (cell-free minus in-cell)
Largest changes in SHAPE reactivity
Protection in cells
Enhanced reactivity in cells
Protein binding
Ratio
Conformational changes
+ΔSHAPE sites
–ΔSHAPE sites
Repeat (E) regions as protein landing pads

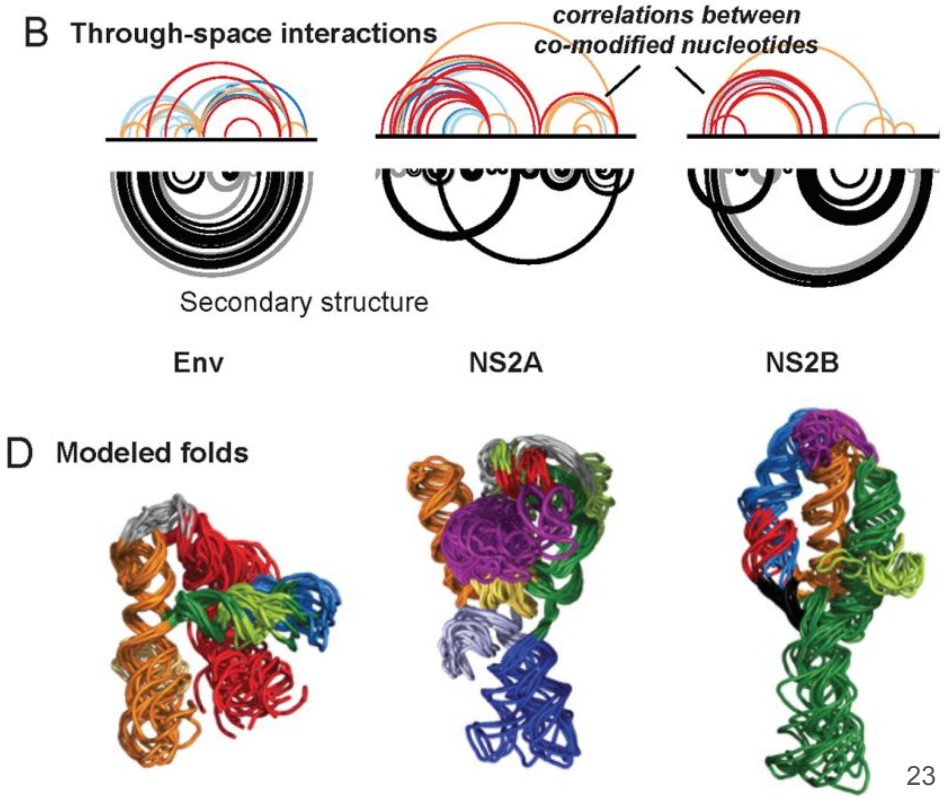
Higher-Order RNA Structures & RING-MaP

Beyond Secondary Structure: Tertiary Interactions

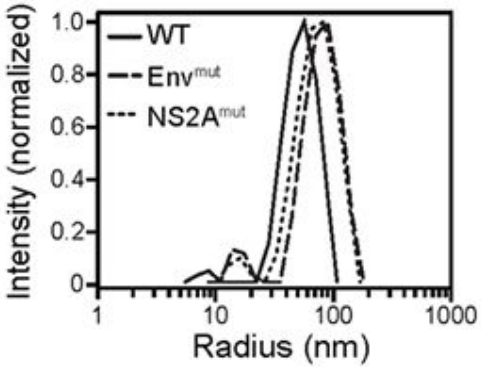
SHAPE identifies base-pairing but **not long-range** interactions

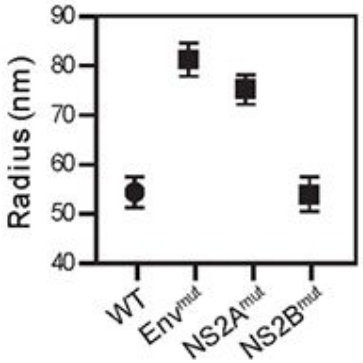
RING-MaP (correlated chemical probing) fills the gap

Examples: Complex Structures in Viral RNAs

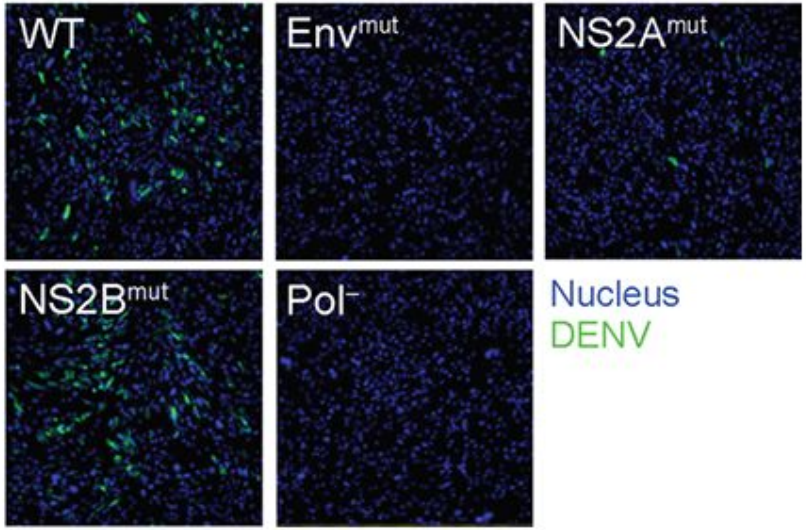


C RNA genome size





Virus fitness



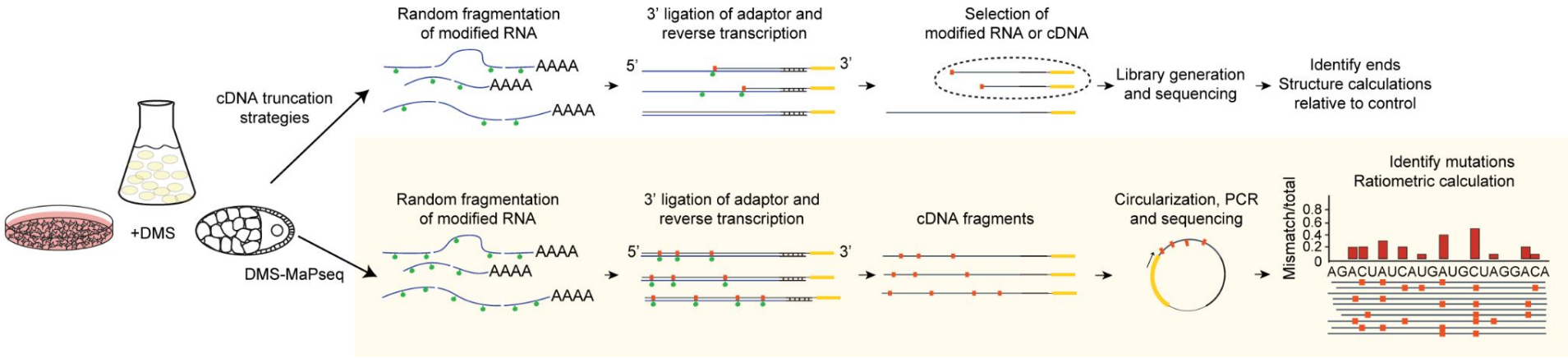
Other probing technique: DMS-MaP

DMS-MaPseq - a chemical probing technique

DMS selectively modifies **unpaired adenines and cytosines**, which are then detected as mutations during reverse transcription and sequencing.

This technique provides high-throughput insights into RNA conformations.

Other powerful method: SHAPE (we talked in Slot 4)



Zubradt, M., Gupta, P., Persad, S., Lambowitz, A. M., Weissman, J. S., & Rouskin, S. (2017). DMS-MaPseq for genome-wide or targeted RNA structure probing in vivo. *Nature methods*, *14*(1), 75-82.

DMS-MaPseq

Breakdown of the name:

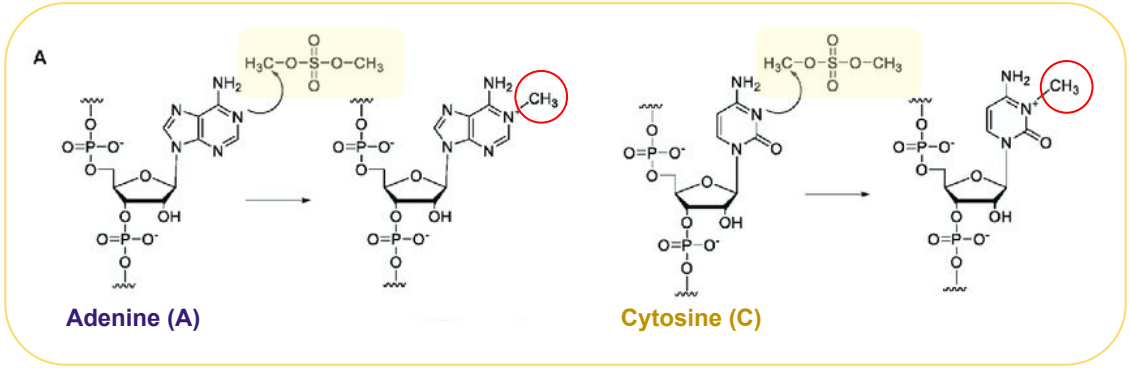
- DMS: dimethyl sulfate (A, C → **A*, C***)
- MaP: mutational profiling
- seq: library sequencing

Ingredients:

- RNA
- TGIRT: thermostable group II reverse transcriptase enzyme (TGIRT-III)
- DMS: dimethyl sulfate

Zubradt, M., Gupta, P., Persad, S., Lambowitz, A. M., Weissman, J. S., & Rouskin, S. (2017). DMS-MaPseq for genome-wide or targeted RNA structure probing in vivo. *Nature methods*, *14*(1), 75-82.

How DMS modified nucleotides



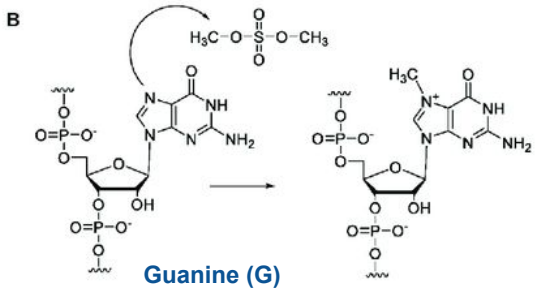
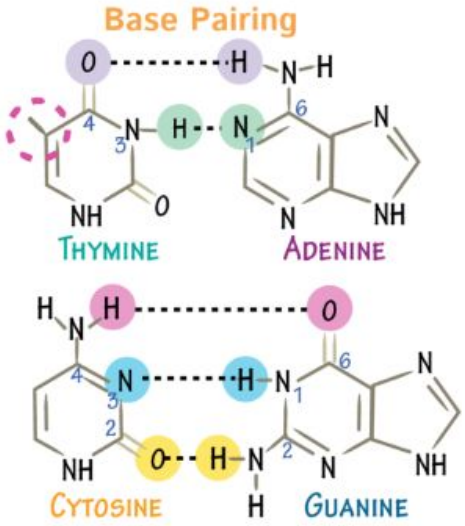
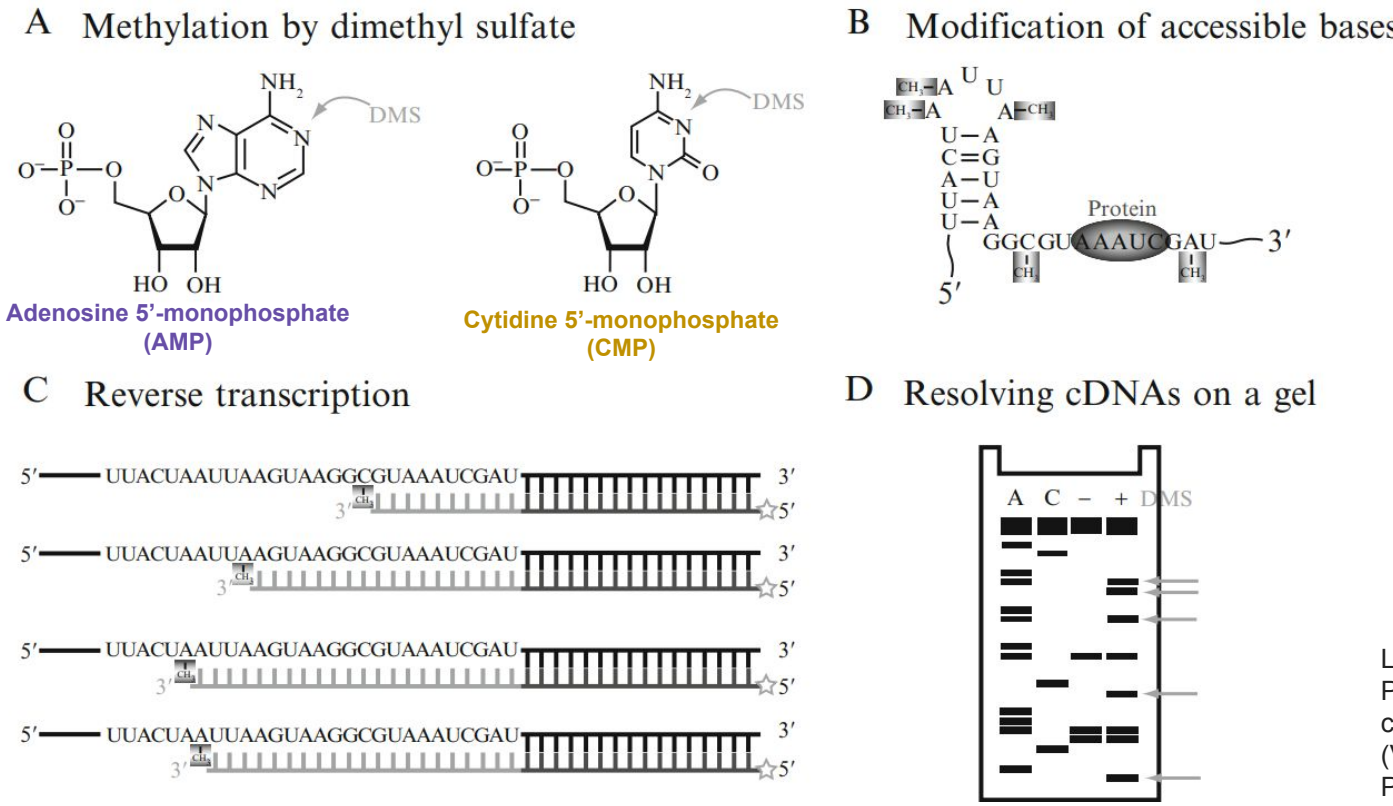


Figure taken from: Waduge, P., Sakakibara, Y., & Chow, C. S. (2019). Chemical probing for examining the structure of modified RNAs and ligand binding to RNA. *Methods*, *156*, 110-120. Link.



“Old-schooled” DMS truncation-based approaches



Liebeg, A., & Waldsich, C. (2009). Probing RNA structure within living cells. In *Methods in enzymology* (Vol. 468, pp. 219-238). Academic Press. Link.

SHAPE-MaPseq vs DMS-MaPseq

Feature	DMS-MaPseq	SHAPE-MaPseq
Chemical Probe Used	Dimethyl Sulfate (DMS)	SHAPE reagents (1M7, NAI, 2A3, etc.)
Modification Targets	Only A & C (N1 of A, N3 of C) methylates specific nitrogen (N) groups in the nucleobases	All four nucleotides (A, C, G, U) Modify 2'-OH of the ribose sugar
Structural Information	Detects base-pairing status	Detects RNA flexibility (measures backbone dynamics)
Works in Living Cells?	Yes	Yes
Can Detect Alternative RNA Conformations?	Yes (with DREEM/DRACO (*slot 5 paper*))	Yes (with DaVinci (*slot 6 paper*))
Process	chemical probing → modification on ribonucleotides → RT (induce mutation) → create cDNA → massive sequencing	

Cao, X., Zhang, Y., Ding, Y., & Wan, Y. (2024). Identification of RNA structures and their roles in RNA functions. *Nature Reviews Molecular Cell Biology*, *25*(10), 784-801.

Conclusion & Future Directions

Key Takeaways

RNA is extensively structured, and SHAPE reveals its function

SHAPE-MaP has revolutionized structural RNA biology

Future focus: large-scale tertiary structure mapping

Discussion

What are other techniques to detect or predict RNA structures?

What are the drawbacks of SHAPE?